

Around the same time, the Americas were setting up their own FDMA network - the AMPS. It had 416 channels (395 for voice and 21 for data) and operated on the 800 MHz band. Though AMPS has been replaced by newer digital technologies such as Digital AMPS (a TDMA network that operates on the existing AMPS bands) and GSM, it still exists as a backup system for voice calls.

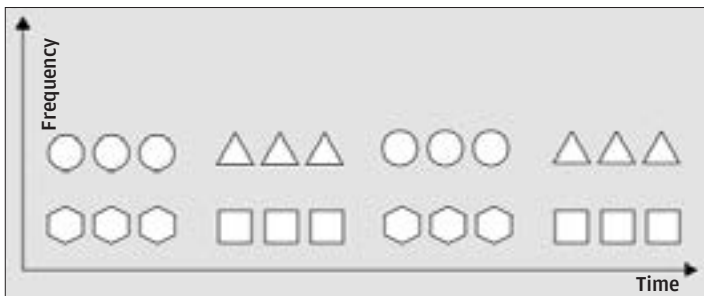
The biggest disadvantage of both these systems was the low call capacity. 395 voice channels on the AMPS network meant that if you were to be the 396th caller in a particular cell, you'd be out of luck!

Another problem with these networks was the lack of security - calls could be overheard with considerable ease, and one could steal a phone's serial code and make illegal calls on the systems.

FDMA networks would soon yield to digital networks, which have more capacity, are more efficient carriers of digital data, and can offer a host of new features such as Web browsing.

Time Division Multiple Access - TDMA

TDMA networks use Time Division Multiplexing to separate callers. Let us consider the case of three callers who have dialled numbers from their cellular phones, and are waiting for their calls to be connected and their conversations to begin. In an FDMA



Our conversations, now in TDMA - notice how we only use two frequency bands this time